Robson Christie

Curriculum Vitae

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PROFILE

Computational physicist and applied mathematician developing stochastic models, numerical methods, and reproducible scientific software. My research combines open quantum systems, rare-event sampling, quantum simulation, and cosmology, taking problems from analytical formulation through implementation and comparison with reference calculations or quantum hardware. Alongside peer-reviewed research, I have designed physics problems to expose weaknesses in frontier AI models and used AI connected to Mathematica through MCP to check research calculations. I am interested in scientific AI, reliable simulation, and tools that connect mathematical reasoning to useful computational results.

EMPLOYMENT HISTORY

Pusan National University Sept 2026–present (FT)
Postdoctoral Researcher

Developing quantum mechanical models of the early universe and black holes, with a focus on Wheeler–DeWitt quantum gravity and open-system approaches.

Mercor Apr 2026–Sept 2026 (FREELANCE)
Physicist

Designed original physics problems for leading AI labs to challenge frontier models and expose failures in scientific reasoning, drawing on theoretical physics and mathematical derivation.

University of Portsmouth March 2024–March 2026 (FT)
Research Fellow

Developed open-system models and simulations across cosmology, quantum computing, and chemical physics. Work includes decoherence and vacuum dynamics during inflation, transition-rate estimation tested on IBM quantum hardware, and electronic-structure and excitonic-coupling calculations for fluorescent proteins.

Imperial College London March 2023–March 2024 (FT)
Postdoctoral Researcher

On EPSRC grant EP/W522673/1, extended transition path and interface sampling to stochastic Schrödinger dynamics, developed rate estimators, and implemented rare-event simulation workflows on HPC clusters using Slurm/PBS. Published in *J. Chem. Phys.* **162**, 114120 (2025).

Imperial College London Sept 2018–March 2023 (FT)
PhD Researcher (Mathematics)

Developed analytical and numerical methods for Lindblad dynamics and stochastic unravellings, including closed-form Gaussian evolution and semi-classical wavepacket methods. PhD awarded March 2023.

TECHNICAL SKILLS

SCIENTIFIC COMPUTING	Python, MATLAB, Mathematica; Qiskit, QuTiP
NUMERICAL METHODS	Stochastic differential equations, Monte Carlo, rare-event sampling, master equations
SOFTWARE & HPC	Git/GitHub, Slurm, PBS; C++ and GLSL project experience
ELECTRONIC STRUCTURE	ORCA, TeraChem; GPU-based QM/MM workflows; basic CUDA
RESEARCH TOOLS	LaTeX; AI-assisted symbolic calculation using Mathematica/MCP

SELECTED SCIENTIFIC SOFTWARE

Quantum dynamics and rare events. Public MATLAB path-sampling code and MATLAB/Mathematica/C++ cosmology solvers, including numerical regression and cross-language checks. Python QM/MM and excitonic-coupling pipeline links electronic structure, molecular dynamics, GPU coupling calculations, and reproducible analysis.

Minkowski Kart. Released C++/GLSL relativistic racing game built on SuperTuxKart with adaptations from OpenRelativity. Developed using AI assistance, combining relativistic visual effects, proper-time telemetry, and physics-inspired gameplay; version 1.92 provides Windows, macOS, Linux, and Android packages. The project documents physical approximations and separates relativistic rendering from arcade mechanics.

EDUCATION

PhD, Mathematics, Imperial College London Sept 2018–March 2023
Advisor: Dr Eva-Maria Graefe. Thesis: *Gaussian dynamics in the Lindblad equation and its stochastic unravellings.*

MSc, Quantum Fields and Fundamental Forces, Imperial College London Sept 2017–Sept 2018
Advisor: Prof Jonathan Halliwell.

BMat&Phys, Mathematics and Physics (First Class Honours), University of Manchester Sept 2014–July 2017

PUBLICATIONS AND PREPRINTS

Peer-reviewed publications

Christie, R., Murray, C., Kim, Y., Joo, J. (2026). Non-Equilibrium Dynamics of the Time-Dependent Excitonic Coupling in Fluorescent Protein Dimers. *J. Phys. Chem. B* **130**(39), 10233–10243. doi:10.1021/acs.jpcb.6c05048; arXiv:2605.00027; code.

Combined coupled-cluster electronic structure, transition-density coupling, and molecular dynamics to quantify thermally averaged excitonic coupling in Venus dimers, with stochastic open-system modelling of subsequent excitation dynamics.

Christie, R., Joo, J., Kaplanek, G., Vennin, V., Wands, D. (2026). Cosmic Lockdown: When Decoherence Saves the Universe from Tunneling. *JCAP* **2026**(09), 125. doi:10.1088/1475-7516/2026/09/125; arXiv:2512.14204; code.

Derived and numerically solved open-system equations for an inflationary scalar field. Non-adiabatic dynamics enhance false-vacuum occupation, while decoherence suppresses subsequent tunnelling. Selected for a SISSA Medialab release.

Christie, R., Bolhuis, P.G., Limmer, D.T. (2025). Transition Path and Interface Sampling of Stochastic Schrödinger Dynamics. *J. Chem. Phys.* **162**(11), 114120. doi:10.1063/5.0246727; arXiv:2411.00490; code.

Extended transition path and interface sampling to stochastic quantum trajectories, enabling rare-transition rate estimation and revealing low-temperature departures from Arrhenius behaviour associated with measurement backaction.

Christie, R., Eastman, J. (2024). Quantum Tunnelling and Thermally Driven Transitions in a Double Well Potential at Finite Temperature. *J. Phys. A: Math. Theor.* **57**(23), 235005. doi:10.1088/1751-8121/ad4b7b; arXiv:2312.12101.

Studied the crossover between thermal activation and quantum tunnelling in a dissipative double well, and developed a Lindblad Brownian-motion model without the zero-temperature singularity.

Christie, R., Eastman, J., Schubert, R., Graefe, E.M. (2022). Quantum-jump vs Stochastic Schrödinger Dynamics for Gaussian States with Quadratic Hamiltonians and Linear Lindbladians. *J. Phys. A: Math. Theor.* **55**(45), 455302. doi:10.1088/1751-8121/ac9d73; arXiv:2203.11530.

Derived closed-form Gaussian evolution for continuous trajectories and a Hagedorn-wavepacket method for jump trajectories, comparing distinct unravellings with the same Lindblad ensemble dynamics.

Preprints and research outputs

Christie, R., Joo, J., Kaplanek, G., Vennin, V., Wands, D. (2026). Quantum Stochastic Inflation. arXiv:2606.12636; source and supporting code.

Formulated stochastic inflation using open quantum systems: derived GKLS dynamics, related Hubble friction and diffusion to one quantum channel, and recovered the classical Fokker–Planck limit.

Christie, R., Baek, K., Bang, J., Joo, J. (2024; revised 2025). Quantum Simulation of Dynamical Transition Rates in Open Quantum Systems. arXiv:2412.17229.

Designed a transition-rate estimation protocol based on independently measurable quantum processes, avoiding full Lindblad evolution; tested on IBM quantum hardware and quantum Brownian-motion simulations.

Christie, R., Trayford, J. (2024). The Sound of Decoherence. arXiv:2412.17045; Python code; Sonification Award Winner 2025.

Mapped quantum density matrices into binaural audio to make tunnelling, thermalisation, decoherence, and recoherence audible.